Hope AI
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Scenario 1:

Flight Delay Analysis

An airline tracks flight delays (in minutes) for 20 flights. Analyze the flight delays to calculate percentiles, detect outliers, and evaluate the overall distribution.

Create a pandas dataframe

Calculate Q1 np.percentile(data, 25) 25th percentile

Calculate Q3 np.percentile(data, 75) 75th percentile

Calculate interquartile range IQR = Q3 – Q1

Calculate the lower and upper bound ranges

Lower bound = Q1 – 1.5 IQR

Upper bound = Q3 + 1.5 IQR

If value is lesser than lower bound then it is an lesser outlier

If value is greater than upper bound then it is an greater outlier

Scenario 2:

Employee Salary Analysis

A company wants to analyze the salary distribution of its employees to understand the central tendency and determine whether the data is skewed.

Determine the lesser and greater outliers first and replace those values with median

Then calculate the mean median mode for the data

If mean>median>mode, Data is right skewed

If mean<median<mode, Data is left skewed

If mean = median = mode then data has no skew

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Scenario 3:

Product Sales Analysis

A retail store records product sales over 15 days. Create a frequency distribution table and visualize the sales data using appropriate charts.

Create a dataframe

Calculate the frequency for all the products from the product column

Calculate the sum of its sales for each product

Use the bar chart to visualize the sales for that particular product

Use line chart to analyze the sales trends for each of the product

Scenario 4:

Student Exam Performance Analysis

A school wants to analyze the exam performance of students across three subjects: Mathematics, Science, and English. How can Data Science concepts be applied to understand their performance?

Calculate total marks per student to see the overall performance

Calculate average to determine which subject the students has scored good marks

Calculate the highest and lowest score of each student to see which student performs well in which subject

Bar chart → compare marks across subjects

Bar chart → compare total marks of students

Line chart → compare subject scores across students

Scenario 5:

Clinical Trial for Diabetes Medication

A pharmaceutical company conducted a clinical trial with two groups: one receiving medication and the other a placebo. Perform a hypothesis test to determine the effectiveness of the medication.

Create a dataset with two groups medication, and placebo

Check the t-test for the each group with other columns

If p<0.05, reject the null hypothesis

then medication is working

If p>0.05 failed to reject null hypothesis accept alternate hypothesis

means the medication not working